



Swathi Sri R 250701769-2025-CSE ▾

S2**Started on** Sunday, 30 August 2026, 7:20 PM**State** Finished**Completed on** Sunday, 30 August 2026, 7:20 PM**Time taken** 38 secs**Marks** 1.00/1.00**Grade** 10.00 out of 10.00 (100%)

Question 1 | Correct | Mark 1.00 out of 1.00

Write a Program to Implement the Quick Sort Algorithm

Input Format:

The first line contains the no of elements in the list-n

The next n lines contain the elements.

Output:

Sorted list of elements

For example:

Input	Result
5	12 34 67 78 98
67 34 12 98 78	

Answer:

```

1  #include <stdio.h>
2  #include <stdlib.h>
3
4  void swap(int *a, int *b) {
5      int temp = *a;
6      *a = *b;
7      *b = temp;
8  }
9
10 int partition(int *arr, int low, int high) {
11     int pivot = arr[high];
12     int i = low - 1;
13
14     for (int j = low; j < high; j++) {
15         if (arr[j] < pivot) {
16             i++;
17             swap(&arr[i], &arr[j]);
18         }
19     }
20     swap(&arr[i + 1], &arr[high]);
21     return (i + 1);
22 }
23
24 void quick_sort(int *arr, int low, int high) {
25     if (low < high) {
26         int pi = partition(arr, low, high);
27         quick_sort(arr, low, pi - 1);
28         quick_sort(arr, pi + 1, high);
29     }
30 }
31
32 int main() {
33     int n;
34     if (scanf("%d", &n) != 1) {
35         return 0;
36     }
37
38     int *arr = (int *)malloc(n * sizeof(int));
39     if (arr == NULL) {
40         return 1;
41     }
42
43     for (int i = 0; i < n; i++) {
44         scanf("%d", &arr[i]);
45     }
46
47     quick_sort(arr, 0, n - 1);
48
49     for (int i = 0; i < n; i++) {
50         printf("%d", arr[i]);
51         if (i < n - 1) {
52             printf(" ");

```

	Input	Expected	Got	
✓	5 67 34 12 98 78	12 34 67 78 98	12 34 67 78 98	✓
✓	10 1 56 78 90 32 56 11 10 90 114	1 10 11 32 56 56 78 90 90 114	1 10 11 32 56 56 78 90 90 114	✓
✓	12 9 8 7 6 5 4 3 2 1 10 11 90	1 2 3 4 5 6 7 8 9 10 11 90	1 2 3 4 5 6 7 8 9 10 11 90	✓

Passed all tests! ✓

Correct

Marks for this submission: 1.00/1.00.

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